

INFORMATION TECHNOLOGY INSTITUTE

LAB 1

gm and *ro* in Triode and Saturation

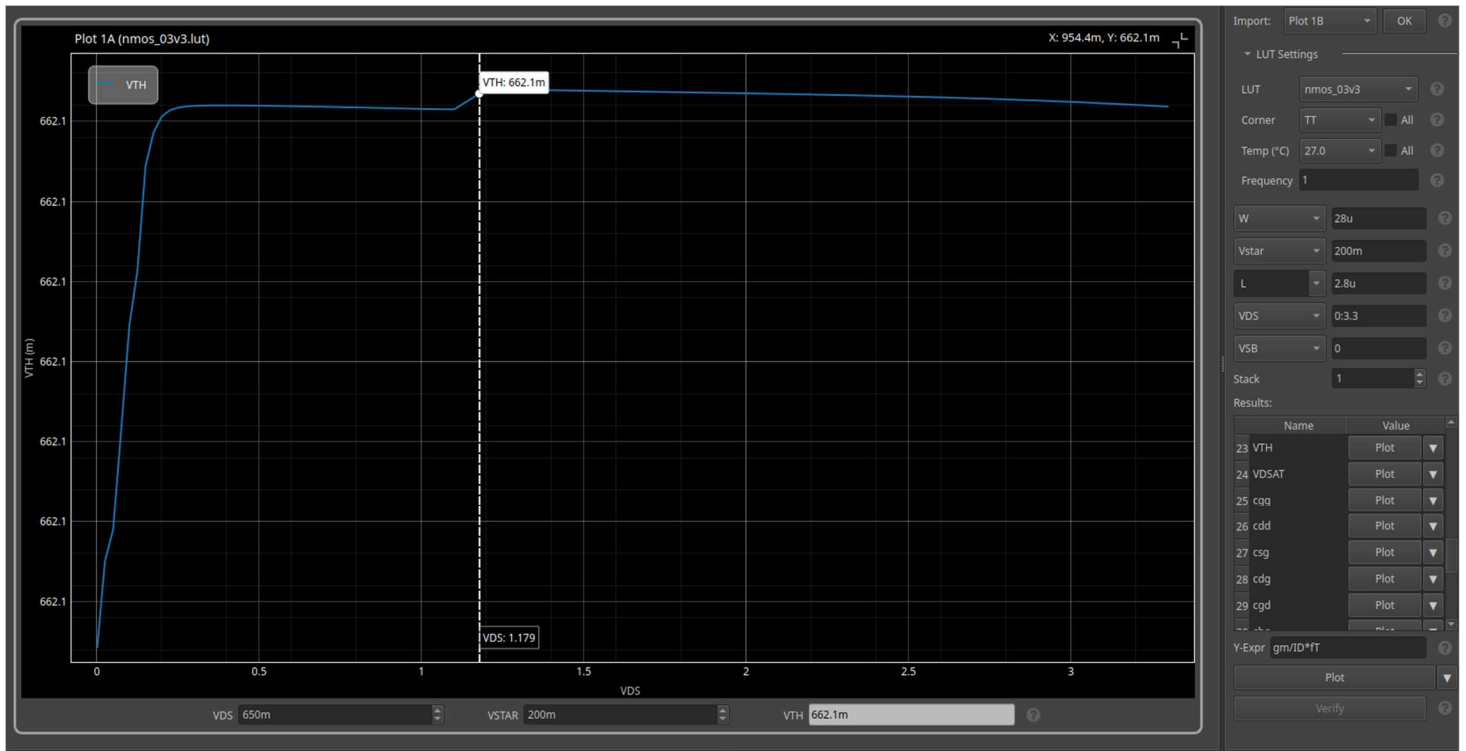




gm and *ro* in triode and saturation

VTH

First we need to calculate the VTH of the nmos

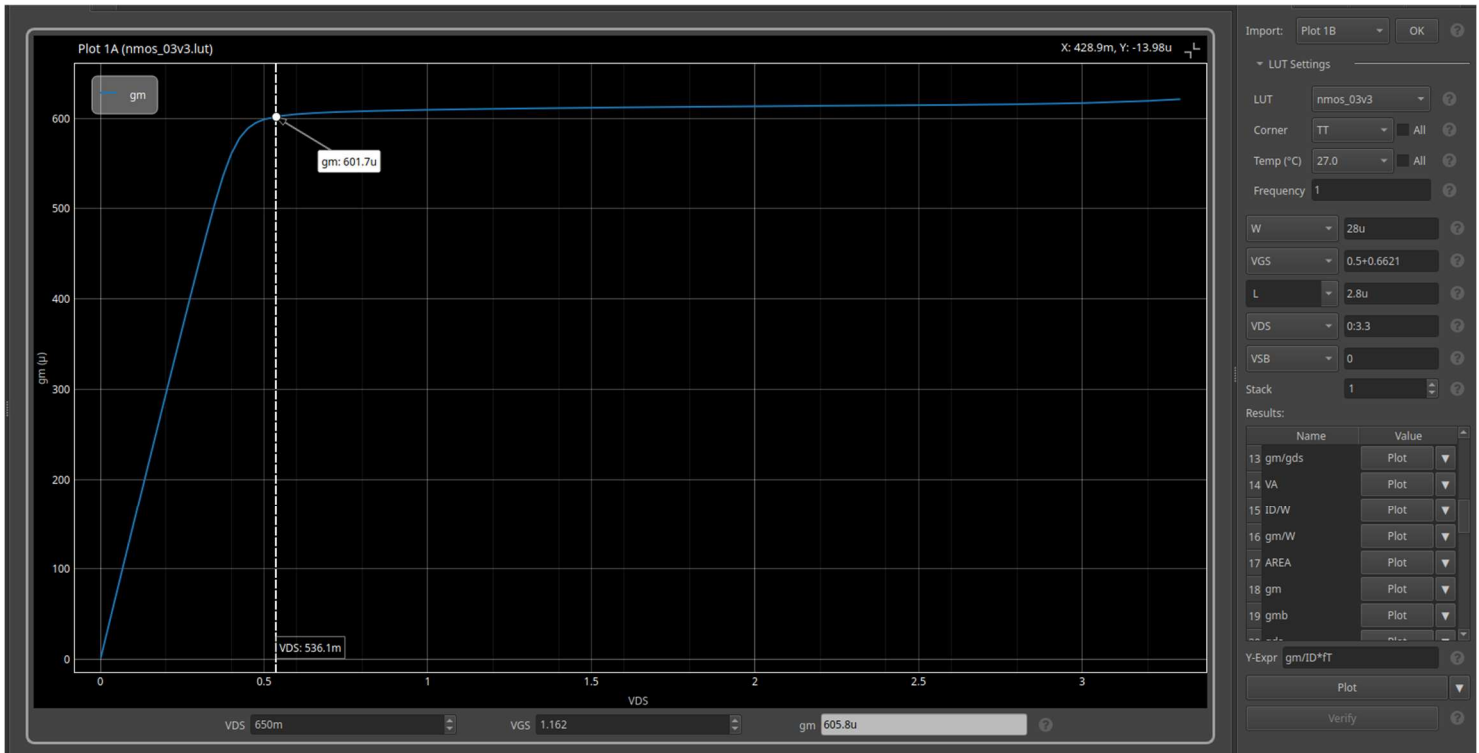


As seen in this graph it should be a constant value of roughly **662.1mV**

So $V_{GS} \approx V_{TH} + 0.5V = 1.1621V$



gm vs Vds



- **Variation of g_m with V_{DS} :**
At first, g_m increases quickly as V_{DS} increases. After that, the curve becomes almost flat at approximately 600 μS .
- **Is the first part linear? Why?**
Yes, it is approximately linear because the transistor is operating in the triode region. In this region:
$$g_m \approx \mu_n C_{ox} \frac{W}{L} V_{DS}$$
Therefore, increasing V_{DS} causes a nearly proportional increase in g_m .
- **Does g_m saturate? Why?**
Yes. Once the transistor enters saturation, the channel becomes pinched off near the drain. The drain current and g_m are then mainly controlled by V_{GS} , so increasing V_{DS} has only a small effect. The slight increase after saturation is caused by nonideal effects such as channel-length modulation and DIBL.



- **At what V_{DS} does the curve saturate?**

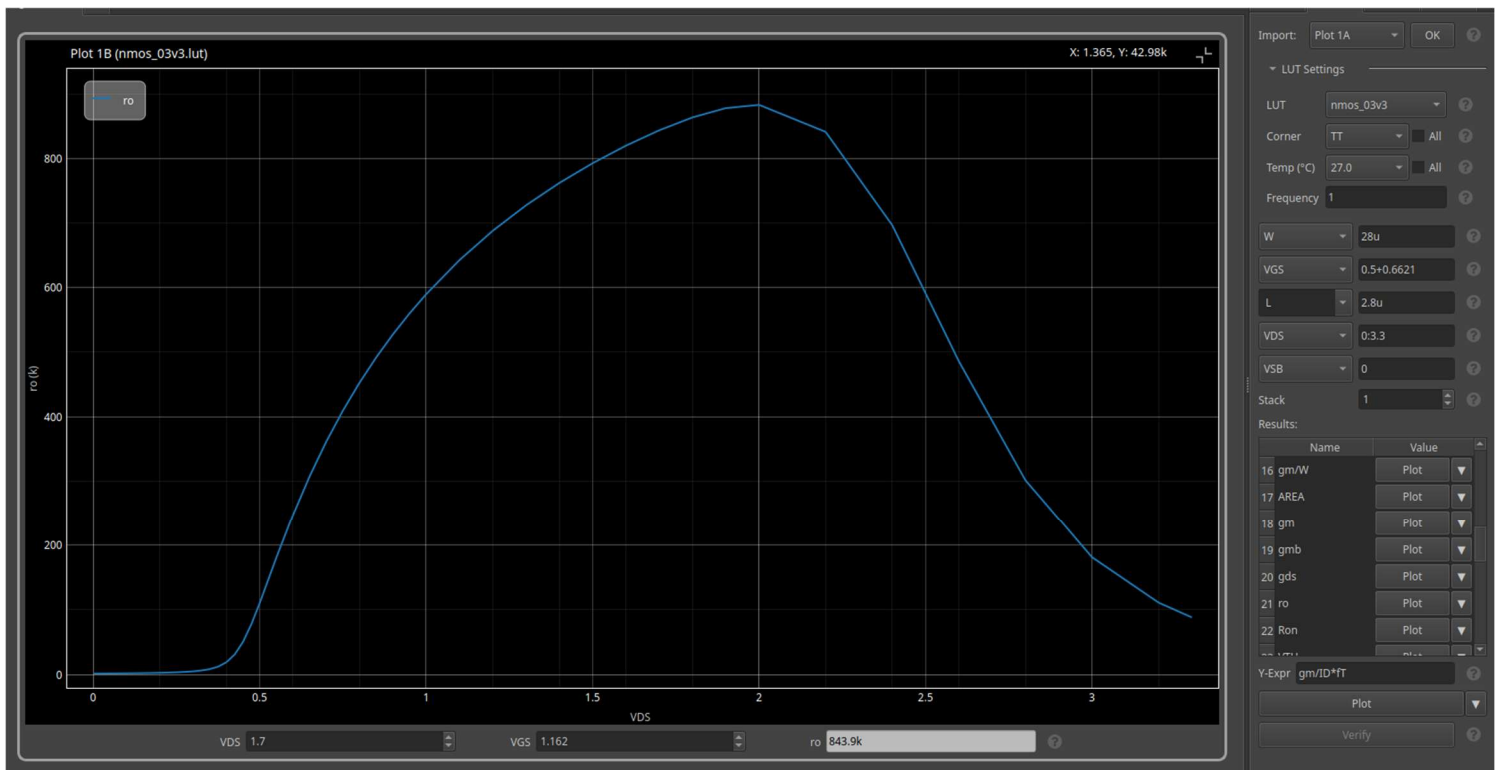
From the graph, saturation begins at approximately: $V_{DS,sat} \approx 0.536 \text{ V}$

In practice, we can consider the saturation transition to occur around 0.5–0.6 V.

- **Where should the transistor operate in an analog amplifier? Why?**

The transistor should operate safely inside the saturation region, slightly above $V_{DS,sat}$, such as the chosen value of $V_{DS} = 0.65 \text{ V}$. In this region, g_m is almost constant, the drain current is mainly controlled by V_{GS} , and the transistor can provide stable voltage gain. Operating slightly above the saturation boundary also provides some margin without wasting too much voltage headroom.

ro vs Vds



- **Variation of r_o with V_{DS} :**

At low V_{DS} , r_o is very small because the transistor is operating in the triode region and I_D strongly depends on V_{DS} . After entering saturation, r_o increases significantly, reaching a maximum of approximately 880–900 kΩ near $V_{DS} = 2 \text{ V}$. It then decreases at higher V_{DS} .



- **Does r_o saturate immediately like g_m ? Why?**

No. Unlike g_m , r_o continues to change after the transistor enters saturation because:

$$r_o = \frac{1}{g_{ds}}$$

As the transistor moves farther into saturation, its $I_D - V_{DS}$ curve initially becomes flatter. Therefore, g_{ds} decreases and r_o continues to increase.

- **Does r_o increase when the transistor is biased further into saturation?**

Yes, initially. According to the graph, r_o increases until approximately $V_{DS} = 2$ V. Beyond this point, r_o decreases because channel-length modulation, DIBL, and other high-field effects increase g_{ds} .

- **Should we operate the transistor at the edge of saturation?**

No. At the edge of saturation, even a small decrease in V_{DS} could push the transistor into the triode region. This would reduce r_o and the amplifier gain and could produce signal distortion.

- **Where should the transistor operate in an analog amplifier? Why?**

The transistor should operate safely inside the saturation region, where r_o is high but before it begins to decrease significantly. From the graph, a reasonable operating range is around:

$$\boxed{V_{DS} \approx 1.7\text{--}2.0 \text{ V}}$$

For example, at $V_{DS} = 1.7$ V, $r_o \approx 844$ k Ω . This point provides high voltage gain, enough margin from the triode region, and reasonable room for the output signal to swing.